MATH 243: Discussion Worksheet 12

Topics: 16.2 Line Integrals (Second Part); 16.3 The Fundamental Theorem for Line Integrals
Please put away all electronic devices, including cell phones and calculators.

1. Evaluate the vector line integral.
   (a) $\displaystyle\int_C y\,dx + 2x\,dy$, where C is parametrized by $x = 3t,\ y = t^2$ with $0 \le t \le 1$.
   (b) $\displaystyle\int_C x\,dx + y\,dy + x^2 y\,dz$, where C is given by $\mathbf{r}(t) = \langle \cos t, \sin t, t\rangle$ with $0 \le t \le \frac{\pi}{2}$.
2. Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F}(x,y) = xy^2\,\mathbf{i} - x^2\mathbf{j}$ and C is given by $\mathbf{r}(t) = t^3\,\mathbf{i} + t^2\,\mathbf{j}$ with $0 \le t \le 1$.
3. Find the work done by the force field
$$\mathbf{F}(x,y) = y^2\,\mathbf{i} - y\sin x\,\mathbf{j}$$
   in moving an object along the curve $x = y^2 + 2$ from $(2,0)$ to $(6,2)$.
4. Determine whether the vector field $\mathbf{F}$ is conservative. If it is, find a function f with $\mathbf{F} = \nabla f$.
   (a) $\mathbf{F}(x,y) = (xy + y^2)\,\mathbf{i} + (x^2 + 2xy)\,\mathbf{j}$
   (b) $\mathbf{F}(x,y) = ye^x\,\mathbf{i} + (e^x + e^y)\,\mathbf{j}$
5. (a) Show that the vector field $\mathbf{F}(x,y) = (1 + xy)e^{xy}\,\mathbf{i} + x^2e^{xy}\,\mathbf{j}$ is conservative and find a potential function f for $\mathbf{F}$.
   (b) Evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$, where C is given by $\mathbf{r}(t) = \cos t\,\mathbf{i} + 2\sin t\,\mathbf{j}$ with $0 \le t \le \frac{\pi}{2}$.
6. Show that the vector line integral $\int_C \sin y\,dx + (x\cos y - \sin y)\,dy$ is independent of path, where C is any piecewise smooth curve from $(2,0)$ to $(1,\pi)$. Then evaluate the integral.
7. Use the Fundamental Theorem of Line Integrals to find the work done by the force field
$$\mathbf{F}(x,y,z) = (ze^y + \cos y)\,\mathbf{i} + (xze^y - x\sin y)\,\mathbf{j} + (xe^y - 2z)\,\mathbf{k}$$
   in moving an object along a piecewise smooth path from $(0,3,2)$ to $(1,0,2)$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Vector Fields. A vector field $\mathbf{F}$ is a **gradient field** if there exists a function f such that

$$\mathbf{F} = \nabla f.$$

In this case, $\mathbf{F}$ is also said to be **conservative**, and f is called a **potential function** for $\mathbf{F}$.

Vector Line Integrals. Let C be a smooth curve given by the vector function $\mathbf{r}$ with $a \leq t \leq b$, and let $\mathbf{F}$ be a continuous vector field. The **vector line integral** of $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\, ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

If $\mathbf{F}(x, y, z) = P(x, y, z)\,\mathbf{i} + Q(x, y, z)\,\mathbf{j} + R(x, y, z)\,\mathbf{k}$, then we also write

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\, dx + Q\, dy + R\, dz$$

and similarly for a curve and vector field in two dimensions.

Work. The **work** done by a force field $\mathbf{F}$ in moving an object along a curve C is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

The Fundamental Theorem for Line Integrals. Let C be a piecewise smooth curve given by $\mathbf{r}(t)$, $a \leq t \leq b$, and suppose that $\mathbf{F} = \nabla f$ for some function f. Assume that $\mathbf{F}$ is continuous. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

Suggested Textbook Problems
Section 16.2 (Second Part): 5-8, 14-24, 29-32, 34, 41-46
Section 16.3: 1-36, 41